

**COMPUTERIZED TESTING
SUPPLEMENT
FOR
RECREATIONAL PILOT
AND
PRIVATE PILOT**

1993

**U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL AVIATION ADMINISTRATION
Flight Standards Service**

SECTIONAL AERONAUTICAL CHART

SCALE 1:500,000

LEGEND

Airports having Control Towers are shown in Blue, all others in Magenta. Consult Airport/Facility Directory (A/FD) for details involving airport lighting, navigation aids, and services.

AIRPORTS

- Other than hard-surfaced runways
- Hard-surfaced runways 1,500 ft. to 8,069 ft. in length
- Hard-surfaced runways greater than 8,069 ft. or some multiple runways less than 8,069 ft.
- Open dot within hard-surfaced runway configuration indicates approximate VOR, VOR-DME, or VORTAC location
- All recognizable hard-surfaced runways, including those closed, are shown for visual identification. Airports may be public or private.

ADDITIONAL AIRPORT INFORMATION

- Private ("Pvt") - Non-public use having emergency or landmark value.
- Military - Other than hard-surfaced. All military airports are identified by abbreviations AFB, NAS, AAF, etc. For complete airport information consult DOD FLIP.
- Helicopter-Selected Public
- Unverified
- Abandoned - paved, having landmark value, 3,000 ft. or greater
- Ultraflight Flight Park Selected
- Services-fuel available and field tended during normal working hours depicted by use of ticks around basic airport symbol. (Normal working hours are Mon thru Fri 10:00 A.M. to 4:00 P.M. local time.) Consult A/FD for service availability at airports with hard-surfaced runways greater than 8,069 ft.
- Relating airport beacons in operation Sunset to Sunrise.

AIRPORT DATA

Box indicates F.A.R. 93

FSS - Flight Service Station
RFSS - Remote Flight Service Station (Canada)
NO SVFR - Fixed-wing special VFR flight is prohibited.
CT - 118.3 - Control Tower (CT) - primary frequency
NFCT - Non-Federal Control Tower
* - Star indicates operation part-time (see tower frequencies tabulation for hours of operation).
① - Indicates Common Traffic Advisory Frequencies (CTAF)
ATIS 123.8 - Automatic Terminal Information Service
ASOS/AWOS 135.42 - Automated Surface Weather Observing Systems. NDB's broadcasting ASOS/AWOS data may not be located at the airport.
UNICOM - Aeronautical advisory station
VFR Advy - VFR Advisory Service shown where ATIS not available and frequency is other than primary CT frequency
285 - Elevation in feet
L - Lighting in operation Sunset to Sunrise
* L - Lighting limitations exist, refer to Airport/Facility Directory.
72 - Length of longest runway in hundreds of feet; usable length may be less.
When facility or information is lacking, the respective character is replaced by a dash. All lighting codes refer to runway lights. Lighted runway may not be the longest or lighted full length. All times are local.

RADIO AIDS TO NAVIGATION AND COMMUNICATION BOXES

- VHF OMNI RANGE (VOR)
- VORTAC
- VOR-DME
- Non-Directional Radiobeacon
- POINT LOMA 302 H+00 & 6m
- Marine Radiobeacon
- Other facilities, i.e., Commercial Broadcast Stations, FSS Outlets, RCO, etc.

122.1R 122.6 123.6

OAKDALE

382 *118.8 OAK

Underline indicates no voice on this freq

* - Operates less than continuous or On-Request.

T - TWB

R - Receive only

H - HIWAS

122.1R

MIAMI

Controlling FSS

CHICAGO CHI

122.1R

Heavy line box indicates Flight Service Station (FSS). Freqs. 121.5, 122.2, 243.0, and 255.4 (Canada - 121.5, 126.7 and 243.0) are normally available at all FSSs and are not shown above boxes. All other freqs. are shown.

For Local Airport Advisory use FSS freq 123.6.

In Canada, all available RFSS frequencies are shown.

Frequencies above this line box are removed to NAVDID site. Other freqs. at controlling FSS may be available as determined by altitude and terrain. Consult Airport/Facility Directory for complete information.

CONTOUR INTERVAL 500 feet

500

HIGHEST TERRAIN elevation is 3818 feet located at 34°52'N - 101°59'W

Critical elevation 4254
Approximate elevation x 3200
Doubtful locations are indicated by omission of the point locator (dot or "x")

AIRPORT TRAFFIC SERVICE AND AIRSPACE INFORMATION

Only the controlled and reserved airspace effective below 18,000 ft MSL are shown on this chart. All times are local.

- Class B Airspace
- Class C Airspace (Mode C See F.A.R. 91.215/AM.)
- Class D Airspace
- Class E Airspace
- Class E Airspace with floor 700 ft. above surface
- Class E Airspace with floor 1,200 ft. or greater above surface that abuts Class G Airspace.

Ceiling of Class D Airspace in hundreds of feet (A minus ceiling value indicates surface up to but not including that value.)

Class E Airspace

Class E Airspace with floor 700 ft. above surface

Class E Airspace with floor 1,200 ft. or greater above surface that abuts Class G Airspace.

2400 MSL

4500 MSL

Class E Airspace low altitude Federal Airways are indicated by center line.

Intersection - Arrows are directed towards facilities which establish intersection.

132° V 69

169

Total mileage between NAVDID's on direct Airways

Prohibited, Restricted, Warning and Alert Areas Canadian Advisory and Restricted Areas

MOA - Military Operations Area

Special Airport Traffic Areas (See F.A.R. Part 93 for details.)

- MODE C (See F.A.R. 91.215/AM.)
- National Security Area
- Terminal Radar Service Area (TRSA)
- MTR - Military Training Routes

OBSTRUCTIONS

- *1000 ft and higher AGL below 1000 ft AGL
- Group Obstruction
- Obstruction with high-intensity lights May operate part-time
- 2049 (1149) - Elevation of the top above mean sea level
- UC - Height above ground
- Under construction or reported position and elevation unverified

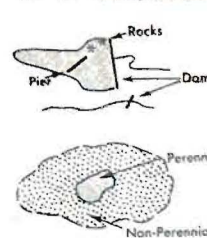
NOTICE: Guy wires may extend outward from structures.

MISCELLANEOUS

- 7°E - Isogonic Line (1990 VALUE)
- Ultralight Activity
- FI - Flashing Light
- Hang Glider Activity
- Marine Light
- Glider Operations
- NAME (Magenta, Blue, or Black)
- Visual Check Point
- Parachute Jumping Area (See Airport/Facility Directory.)

TOPOGRAPHIC INFORMATION

- Roads
- Road Markers
- Railroad
- Bridges And Viaducts
- Power Transmission Line
- Aerial Cable
- Landmark Feature - stadium, factory, school, golf course, etc.
- Outdoor Theatre
- Lookout Tower P-17 (Site Number) 618 (Elevation Base of Tower)
- CG Coast Guard Station
- Race Track
- Tank-water, oil or gas
- Oil Well - Water Well
- Mines And Quarries
- Mountain Pass
- 11823 (Elevation of Pass)



ATTENTION

THIS CHART CONTAINS MAXIMUM ELEVATION FIGURES (MEF). The Maximum Elevation Figures shown in quadrangles bounded by ticked lines of latitude and longitude are represented in THOUSANDS and HUNDREDS of feet above mean sea level. The MEF is based on information available concerning the highest known feature in each quadrangle, including terrain and obstructions (trees, towers, antennas, etc.)

Example: 12,500 feet

MILITARY TRAINING ROUTES (MTRs)

All IR and VR MTRs are shown, and may extend from the surface upwards. Only the route centerline, direction of flight along the route and the route designator are depicted - route widths and altitudes are not shown. Since these routes are subject to change every 56 days, and the charts are reissued every 6 months, you are cautioned and advised to contact the nearest FSS for route dimensions and current status for those routes affecting your flight. Routes with a change in the alignment of the charted route centerline will be indicated in the Aeronautical Chart Bulletin of the Airport/Facility Directory. Military Pilots refer to Area Planning AP/18 Military Training Route North and South America for current routes.

LEGEND 1.—SECTIONAL AERONAUTICAL CHART.

INK SA 1854 CLR 15 106/ 77/63/1112G18/000
 BOI SA 1854 150 SCT 30 181/62/42/1304/015
 LAX SA 1852 7 SCT 250 SCT 6FK 129/60/59/2504/991
 MDW RS 1856 -X M7 OVC 11/2R+F 990/63/61/3205/980/RF2 RB12
 JFK RS 1853 W5 X 1/2F 180/68/64/1804/006/R04RVR22V30 TWR VSBY 1/4

FIGURE 12.—Surface Aviation Weather Report.

This is a telephone weather briefing from the Dallas FSS for a local operation of gliders and lighter-than-air at Caddo Mills, Texas (about 30 miles east of Dallas). The briefing is at 13Z.

"There are no adverse conditions reported or forecast for today.

"A weak low pressure over the Texas Panhandle and eastern New Mexico is causing a weak southerly flow over the area.

"Current weather here at Dallas is clear, visibility 12 miles, temperature 70, dewpoint 48, wind south 5 knots, altimeter 29 point 78.

"By 15Z we should have a few scattered cumuliform clouds at 5 thousand AGL, with higher scattered cirrus at 25 thousand MSL. After 20Z the wind should pick up to about 15 knots from the south.

"The winds aloft are: 3 thousand 170 at 7, temperature 20; 6 thousand 200 at 18, temperature 14; 9 thousand 210 at 22, temperature 8; 12 thousand 225 at 27, temperature 0; 18 thousand 240 at 30, temperature -7."

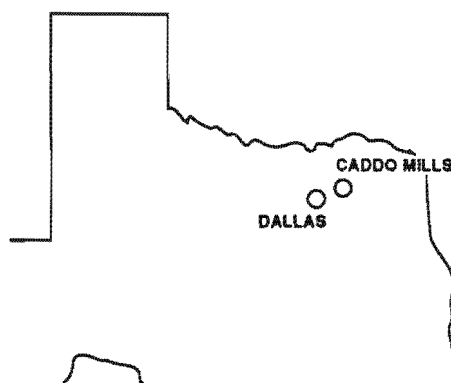


FIGURE 13.—Telephone Weather Briefing.

UA /OV OKC-TUL /TM 1800 /FL 120 /TP BE90 /SK 018 BKN 055 /
/072 OVC 089 /CLR ABV /TA -9/WV 0921/TB MDT 055-072 /IC LGT-MDT
CLR 072-089.

FIGURE 14.—Pilot Weather Report.

OK FT 011447

GAG FT 011515 100 SCT 250 SCT 2610. 16Z 60 SCT C100 BKN 3315G22 CHC C50 BKN
5TRW. 01Z 250 SCT 3515G25. 09Z VFR WIND..

HBR FT 011515 C120 BKN 250 BKN 3010. 17Z 100 SCT C250 BKN 3215G25 CHC C30 BKN
3TRW. 00Z 250 SCT 3515G25. 09Z VFR WIND..

MLC FT 011515 C20 BKN 1815 BKN OCNL SCT. 20Z C30 BKN 1815G22 CHC C20 BKN
1TRW. 03Z C30 BKN 2015 CHC C7 X 1/2TRW+G40. 09Z MVFR CIG TRW..

OKC FT 011515 C12 BKN 140 BKN 1815G28 LWR BKN V SCT. 18Z C30 BKN 250 BKN
2315G25 LWR BKN OCNL SCT CHC C7 X 1/2TRW+G40. 21Z CFP 100 SCT C250 BKN
3315G25 CHC C30 BKN 5TRW-. 02Z 100 SCT 250 SCT 3515G25. 09Z VFR WIND..

PNC FT 011515 C100 BKN 250 BKN 1810. 16Z CFP 20 SCT C100 BKN 3115 SCT V BKN. 00Z
250 SCT 3515G25. 09Z VFR WIND..

TUL FT 011515 C20 BKN 1915G22. 19Z C30 BKN 1815G25 CHC 3TRW. 23Z CFP C100 BKN
250 BKN 3215G25 CHC C30 BKN 5TRW. 09Z VFR WIND..

FIGURE 15.—Terminal Forecast.

DFWH FA 041040
 HAZARDS VALID UNTIL 042300
 OK TX AR LA TN MS AL AND CSTL WTRS
 FLT PRCTNS. . .TURBC. . .TN AL AND CSTL WTRS
 . . .ICG. . .TN
 . . .IFR. . .TX
 TSTMS IMPLY PSBL SVR OR GTR TURBC SVR ICG AND LLWS
 NON MSL HGTS NOTED BY AGL OR CIG
 THIS FA ISSUANCE INCORPORATES THE FOLLOWING AIRMETS STILL IN
 EFFECT. . .NONE.

DFWS FA 041040
 SYNOPSIS VALID UNTIL 050500
 AT 11Z RDG OF HI PRES ERN TX NWWD TO CNTRL CO WITH HI CNTR
 OVR ERN TX. BY 05Z HI CNTR MOVS TO CNTRL LA.

DFWI FA 041040
 ICING AND FRZLVL VALID UNTIL 042300
 TN
 FROM SLK TO HAT TO MEM TO ORD TO SLK
 OCNL MDT RIME ICGIC ABV FRZLVL TO 100. CONDS ENDING BY 17Z.
 FRZLVL 80 CHA SGF LINE SLPG TO 120 S OF A IAH MAF LINE.

DFWT FA 041040
 TURBC VALID UNTIL 042300
 TN AL AND CSTL WTRS
 FROM SLK TO FLO TO 90S MOB TO MEI TO BUF TO SLK
 OCNL MDT TURBC 250-380 DUE TO JTSTR. CONDS MOVG SLOLY EWD
 AND CONTG BYD 23Z.

DFWC FA 041040
 SGFNT CLOUD AND WX VALID UNTIL 042300. . .OTLK 042300-050500
 IFR. . .TX
 FROM SAT TO PSX TO BRO TO MOB TO SAT
 VSBY BLO 3F TIL 15Z.
 OK AR TX LA MS AL AND CSTL WTRS
 80 SCT TO CLR EXCP VSBY BLO 3F TIL 15Z OVR PTNS S CNTRL TX.
 OTLK. . .VFR.
 TN
 CIGS 30-50 BKN 100 VSBYS OCNLY 3-5F BCMG AGL 40-50 SCT TO
 CLR BY 19Z. OTLK. . .VFR.

FIGURE 16.—Area Forecast.

Appendix 3

FD WBC 151745
 BASED ON 151200Z DATA
 VALID 1600Z FOR USE 1800-0300Z. TEMPS NEG ABV 24000

FT	3000	6000	9000	12000	18000	24000	30000	34000	39000
ALS			2420	2635-08	2535-18	2444-30	245945	246755	246862
AMA		2714	2725+00	2625-04	2531-15	2542-27	265842	256352	256762
DEN			2321-04	2532-08	2434-19	2441-31	235347	236056	236262
HLC		1707-01	2113-03	2219-07	2330-17	2435-30	244145	244854	245561
MKC	0507	2006+03	2215-01	2322-06	2338-17	2348-29	236143	237252	238160
STL	2113	2325+07	2332+02	2339-04	2356-16	2373-27	239440	730649	731960

FIGURE 17.—Winds and Temperatures Aloft Forecast.

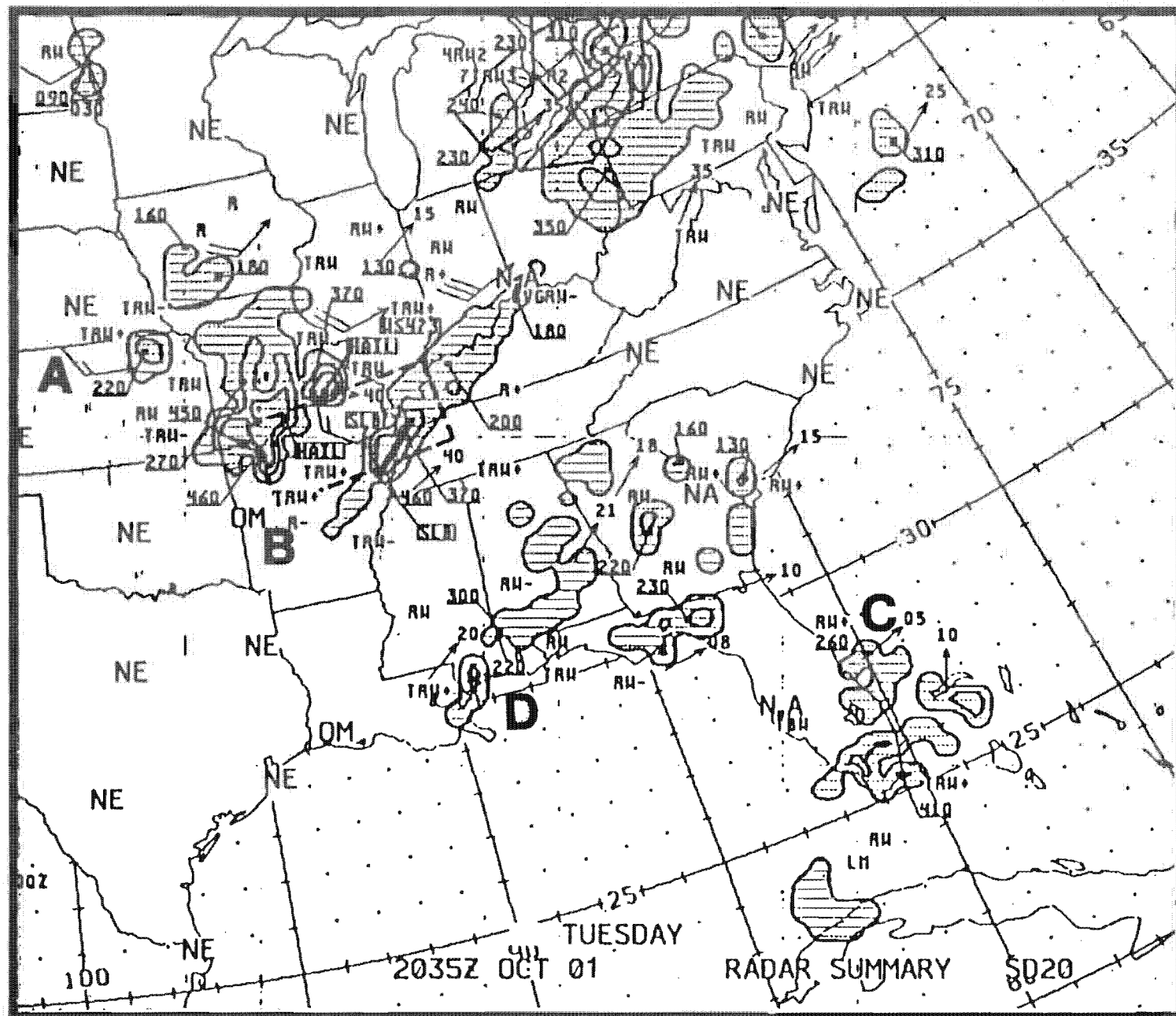


FIGURE 19.—Radar Summary Chart.

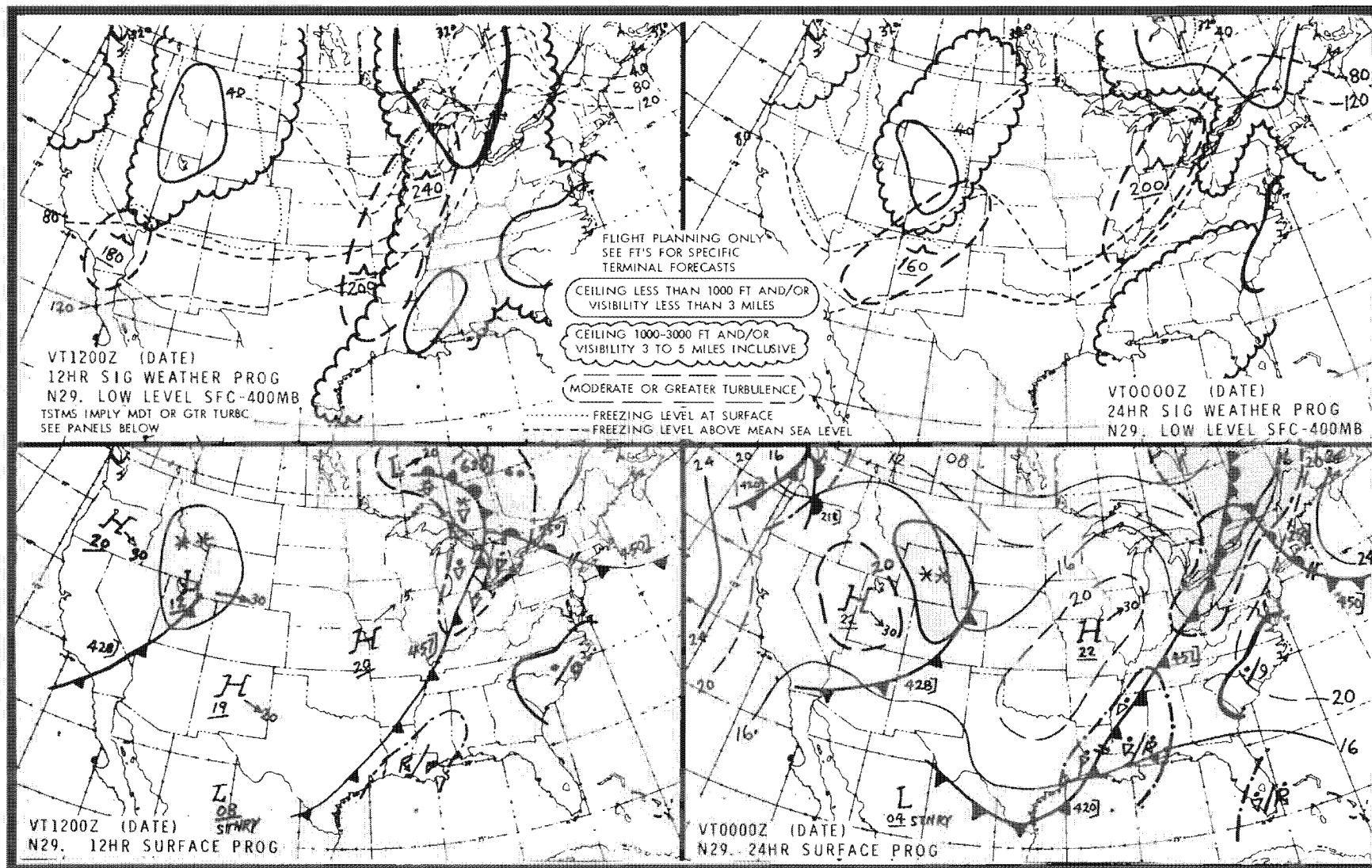


FIGURE 20.—Significant Weather Prognostic Chart.